Science Lab: development plan

Vermont Literacy Network student dashboard · October 15, 2026

Build Queue 198 and 199 (2026-10-15). Her words, verbatim: "give me the dev plan for science lab. claudia we need to improve this. remove the potions in supernature button from this"

Where it lives

- On the student computer: 🔬 Science Lab (`/student/science-lab`). Code: `src/routes/student/ScienceLab.tsx` ; chemistry and recipes: `src/games/scienceLab/lab.ts` ; Atom Builder: `src/games/scienceLab/AtomBuilder.tsx` ; periodic table data and element uses: `src/games/scienceLab/elements.ts` .
- Saves per student in the `lab:<studentId>` row (no new SQL): bottles unlocked, recipes found, atoms built.
- Full history: dev plan Part A, A56, and Build Queue items 108, 156, 165, 179 and 198 to 200.

What is built today

- **A 2D lab, top to bottom:** the periodic table board on the wall, a bottle shelf with ◀ ▶ arrows, and a bench with 3 glass beakers. Tap a beaker, then tap a bottle to pour; the bottle tilts, a stream pours, the liquid rises. Each beaker has Pour into..., Heat and Empty. A beaker holds 5 pours.
- **36 real recipes:** 31 mixing recipes and 5 burner recipes (boiling water, salt from salt water, caramel, blue crystals, glowing iron sulfide). Fizzing volcano, sodium and potassium in water (BOOM), hydrogen and oxygen (POP), red cabbage indicator, rusting iron, elephant toothpaste, thermite and more. Things explode only when the real reaction would; mixes that do not react blend colors and say "No reaction"; helium explains it is a noble gas.
- **Learned over time:** the shelf starts with 12 bottles, and discoveries unlock 18 more (30 bottles). Elements on the shelf and atoms built in the Atom Builder light up on the periodic table.
- **The periodic table:** all 118 elements in the real layout, colored by family. Tap the board to open it big with a color key; tap any element for its card: number, family, a fact, a real photo, "Used for" (55 elements) and how to unlock it.
- ⚛️ **Atom Builder:** + and − for protons, neutrons and electrons with filling shells; says which element, its charge, its mass number and, for hydrogen to calcium, whether the nucleus is stable.
- 📓 **Science Journal:** every discovered recipe with ingredients, equation and a one-line fact, plus a "Not discovered yet" slot for each one left.
- **Sandbox only:** no questions, no money, no timer. Sounds made in the browser with a mute button. Reduced motion turns off the shake and flashes.
- **Moved out (2026-10-14):** the Magic mode, Potion Lab and Spellbook now live in Supernature's Cauldron, which shares this chemistry and these discoveries.
- **Shipped today (item 200):** the "🔮 Potions are in Supernature" button is gone from the top bar.

Claudia's review: what is holding it back

Checked against the platform standard (iPad first, text to speech on every screen, high-support learners, clear next step, play first with the learning inside).

1. **The bench is below the fold.** In her screenshot the periodic table fills the screen and the beakers and shelf are cut off at the bottom. The thing students came to do (pour and mix) is the thing they cannot see. On an iPad in landscape it is worse. The table should be small on the wall (or a button), and the bench should fill the screen.
2. **No clear next step.** It is a pure sandbox. Students who know what to try have fun; students with high support needs see 12 bottles and 3 beakers and stop. There is nothing that says "try this", no goal, and no help that fades.
3. **No prediction.** Real science is predict, try, explain. Right now a student pours and sees what happens, but is never asked to guess first, so the reaction is a surprise instead of a test of an idea. (Supernature's Cauldron already does guess and check, item 179.)
4. **Little read-aloud.** The page has the shared 🔊 Listen button, but the discovery card, the element cards and the journal do not read themselves aloud, and the reaction words (BOOM, POP) are not spoken. Most of the lab's teaching lives in those cards.
5. **The journal is a long list.** 36 entries in one scrolling box, with locked ones that only say "3 ingredients". Supernature's Element Book (a page-turning book, one thing per spread) is a better model and would make the two apps feel like one family.
6. **Safety is never mentioned.** Sodium in water, thermite and hydrogen with chlorine are real and dangerous. Each dangerous reaction should say plainly that it is only done by scientists with protection, never at home.
7. **Two taps to pour, with no drag.** Tap a beaker, then a bottle, works on an iPad, but students expect to drag a bottle to a beaker. Adding drag (keeping tap) makes it feel like a real bench.
8. **Leftover Magic code.** The page still carries the old magic switches (always off now). Cleaning it out makes future work faster and safer.
9. **No link to class work.** No questions, no money, nothing in her reports. That is fine for a sandbox, but there is no way for her to see what a student discovered.

Proposed improvements, in the order Claudia would build them

Phase 1: make the bench the star (small, ready to build).

- Bench first: beakers and shelf fill the screen on an iPad in both orientations; the periodic table becomes a slim strip (or a 🧪 Table button) that opens big when tapped. Nothing important below the fold.
- 🔊 on every card: discovery card, element card, journal page and Atom Builder card read aloud with the human voice; BOOM and POP are spoken too.
- Safety lines on the dangerous recipes ("Only scientists with goggles and a fire blanket do this. Never at home.").
- Drag a bottle onto a beaker (tap still works); pour from beaker to beaker by dragging too.
- Remove the leftover Magic code.

Phase 2: a clear next step (the biggest win for her students).

- **Try This cards:** a small card on the bench with one experiment to try next ("Can you make something fizz? Try vinegar and baking soda."). Help fades, like the Cauldron: first a riddle, then the right bottles glow, then the exact bottle glows.

- **Predict first:** before a reaction runs, three or four picture choices (Fizz, Boom, Color change, Nothing). After it runs: "You said fizz. It fizzed! That is because..." A wrong guess is never punished; it just shows what really happened.
- **Science Journal as a book:** one recipe per spread, with the picture, ingredients, equation, fact, safety line and 🔊; locked pages show a silhouette and a Try This hint.

Phase 3: grow the science (pick any).

- **Measure and change:** a thermometer and temperature on the burner; pH paper next to the cabbage indicator; "more or less" (a pinch or a scoop) for a few reactions where amount matters.
- **States of matter station:** freeze, melt, boil and condense water and other things (ice, steam, dry ice) with particle views.
- **Atom Builder to molecules:** snap atoms together to build water, carbon dioxide, salt and oxygen gas, tying the Atom Builder to the bench.
- **Lab badges and a teacher view:** badges for families of discoveries (Acids and bases, Explosions, Burner chef), and a per-student list in her reports of what they discovered.
- **Optional questions:** her question sets unlock new bottles or Try This cards (only if she wants questions here; it is sandbox only today).

Questions for her before building

1. Phase 1 first, then Phase 2? (Claudia's suggestion: yes. Phase 1 is fixes; Phase 2 is the real improvement.)
2. Periodic table on the bench: a slim strip on the wall, or only a 🧪 Table button? (Suggestion: a slim strip that opens big on tap, so it stays "a board in the background" as in her first plan.)
3. Predict first: every time, or only on recipes not found yet? (Suggestion: only the first time, so repeats stay quick.)
4. Should the Science Lab ask questions or pay Class Cash, or stay a free sandbox? (Suggestion: stay a sandbox, with badges and a teacher view.)
5. Which Phase 3 ideas, if any?